Weather sensitivity of bike-share usage: a hierarchical Bayesian plan for 25 cities

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Abstract

Weather is one of the few influences on bike-share usage that an operator cannot control. We assemble a reproducible daily panel from the public release behind Bean et al. [2021a]: 29 dock-based systems with 8,760 hourly records each, reduced to 10,578 city-days. The primary analysis keeps the 25 systems that record hires on at least 99% of their analysis days, giving 9,120 city-days. Median daily hires span 6,618-fold across these cities, which is the empirical reason for treating cities as a hierarchy. Binned within-city averages show a thermal response peaking near 27.5°C and then falling, and a rainfall response that declines monotonically to about 41% below the dry-day bin. We close with the hierarchical Student-t regression and inference plan for the final project.

1 Introduction

Bike-share is part of the transport mix in many cities, and usage responds to weather in ways operators can plan around.

The reference study [Bean et al., 2021a] compares forty cities across climate zones. It fits a frequentist generalized additive model to each city separately and reports two robust patterns: hires rise with the Universal Thermal Climate Index (UTCI) up to roughly 27–28°C and then decline, and rainfall is associated with lower use. That design gives a clear within-city picture but leaves two questions open. Modelling each city in isolation cannot quantify how much cities differ, because no quantity in the model describes the spread of the coefficients. And small systems are estimated noisily: median daily hires in our panel range from 11.5 in Créteil to $76{,}103$ in Paris, so low-volume systems carry far less signal per day.

We therefore recast the comparison as a hierarchy. Each city keeps its own intercept, weather coefficients, weekend coefficient and seasonal coefficients, but those city parameters share a normal population distribution. This answers three questions that the midterm is designed to set up.

> **Research questions.** (1) How does thermal comfort relative to the local study-period average relate to daily hires? (2) How much do cities differ in their rainfall response? (3) Do a city's thermal and rainfall background explain part of that between-city variation?

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Table 1: Frozen daily panel and primary sample. The full 29-city panel is kept for the sample-range sensitivity analysis planned for the final project.

Item	Value	Item	Value
Dock-based cities	29	Primary cities	25
Hourly records	254,040	Primary city-days	9,120
City-days after filtering	10,578	Zero-hire days, primary	4
Date range	2016-07 – 2017-12	Dry-day share	21.5%
Countries or regions	10	Trewartha codes	6

This report documents the data, the cleaning decisions and the exploratory evidence, and states the model the final project will fit. All statements describe statistical association; the data cannot identify a causal effect.

2 Data

Source and scope. The data are the public release accompanying Bean et al. [2021a], distributed through Mendeley Data [Bean et al., 2021b] under CC BY 4.0 (DOI `10.17632/2nxpvtz935.1`, Version 1). Hourly hires are estimated from per-minute station inventory snapshots and may carry a 0.5 fractional part; UTCI and precipitation come from the Copernicus Climate Data Store. The archive carries metadata and weather arrays for forty cities but hourly stock data for thirty; the paper's R code designates metadata rows 7–34 and row 36 as the 29 dock-based systems, and we keep that scope. Archive and paper are stored with SHA-256 digests re-checked on every run.

Hourly to daily. Each city contributes exactly 8,760 stock rows. The source file contains three repeated city–date–hour keys: Brisbane and Dublin on 2017-10-29 at 02:00, and Melbourne on 2017-04-02 at 02:00. The Dublin and Melbourne dates coincide with daylight-saving transitions, whereas Brisbane does not observe daylight saving, so the cause of its duplicate is left unspecified. Following the paper's rule we preserve row order and map the 8,760 rows onto consecutive UTC hours from the first source date rather than merging the repeated labels. The hourly series is then converted to each city's local time zone and aggregated by local date: hires and precipitation are summed, UTCI contributes its daily mean, minimum and maximum. A local day enters the analysis only if it lies inside the city's nominal study period, contains 23 to 25 hourly rows, has at least 20 valid UTCI hours, and has complete precipitation. Of 10,613 candidate city-days, 35 fail at least one rule, leaving 10,578; the reasons overlap: 28 lie outside the nominal period, 34 outside the 23–25 hour count and 33 fall short of 20 UTCI hours.

Primary sample. Of the 575 zero-hire days in the full panel, 571 sit in Göteborg, Kazan, Lillestrøm and Vilnius; long zero runs can reflect seasonal operation, station recording state or genuinely no use, and the three are not separable. We therefore restrict the primary analysis in advance to cities recording hires on at least 99% of their analysis days, keeping 25 cities and 9,120 city-days and leaving four zero-hire days. The full 29-city panel is retained in the same file behind a `continuous_system` flag. The processed panel has no missing cells, no negative values and a unique city–date key; re-running the preparation script reproduces it byte-for-byte. Table 1 summarises the panel.

3 Exploratory analysis

Everything below is computed from the frozen panel with no model and no imputation. Averages are taken inside cities first and then across cities, so a single large system cannot dominate a curve.

City scale. Figure 1a orders the 25 primary cities by median daily hires, a 6,618-fold spread. Figure 1b shows the continuous-system rule: the four excluded cities record positive hires on between 35% and 92% of analysis days, while every primary city sits above 99%.

Thermal response. Figure 2a bins city-days by daily mean UTCI in 5°C intervals and plots within-city centred $\log(1 + \text{hires})$ against the bin centre, with the 25th to 75th percentile band across cities. The curve reaches its global maximum of $+0.26$ near 27.5°C, against -0.26 in the

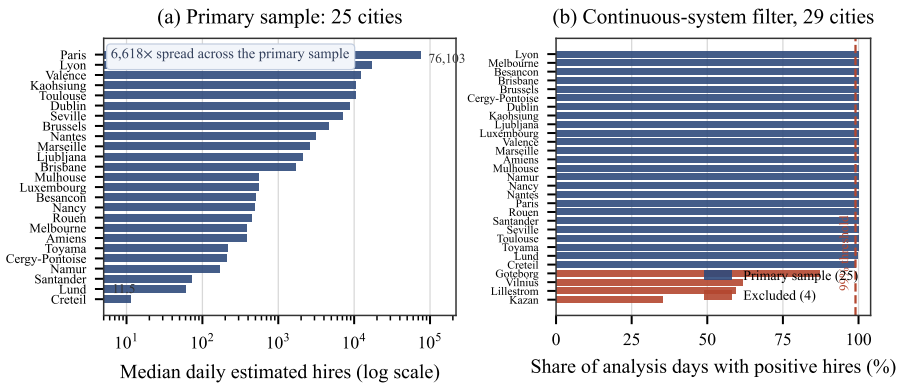


Figure 1: (a) Median daily hires for the 25 primary cities on a log scale. (b) Share of analysis days with positive hires for all 29 dock-based cities, with the 99% inclusion threshold.

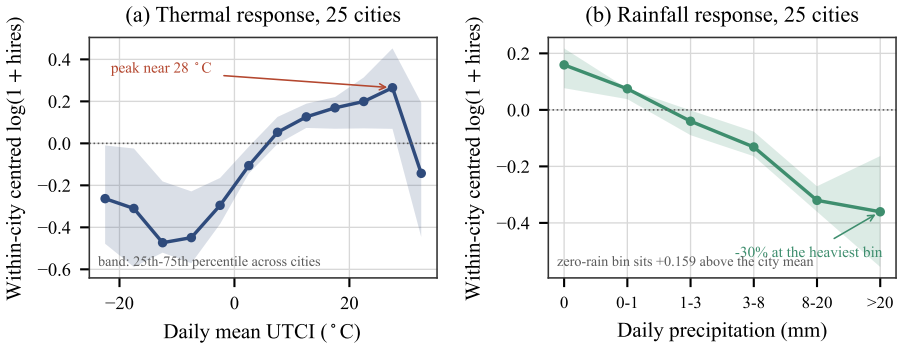


Figure 2: Within-city centred $\log(1 + \text{hires})$ against (a) daily mean UTCI and (b) daily precipitation, for the 25 primary cities. Points are city-equal bin averages; the shaded band spans the 25th to 75th percentile across cities.

coldest populated bin (-22.5°C) and -0.14 in the hottest bin (32.5°C); the three coldest bins are not ordered monotonically. The peak agrees with the 27–28°C of Bean et al. [2021a], and the drop to the hottest bin is 0.41 log units. The band across cities is 0.38 log units wide at the peak, which is the between-city variation that questions (2) and (3) target.

Rainfall response. Figure 2b bins city-days by daily precipitation. The dry bin sits $+0.16$ above the city mean and the response declines monotonically across the six bins, reaching -0.36 above 20 mm, which is 0.52 log units and about 41% below the dry bin on the hires scale. That contrast is descriptive: it compares two bins of the pooled curve with no adjustment for season, city or the other covariates. Rainfall has a consistent sign, but the figure shows only the pooled shape; how much it varies across cities is what the hierarchical model will estimate.

Season and week. Figure 3a shows the seasonal profile, peaking in mid-October and troughing at the end of December with an amplitude of 0.84 log units, so the annual cycle is larger than the whole thermal effect. Figure 3b ranks cities by the weekend-minus-weekday difference: 23 of 25 are negative, with a mean of -0.55, and Melbourne is the only clearly positive city at $+0.27$.

Implications for the specification. Three features shape the model: the thermal response is non-monotone, so a quadratic term is needed; between-city variation is visible in every panel, so city coefficients must be free to differ; and the seasonal amplitude exceeds the weather effects, so the annual cycle must be absorbed before weather coefficients can be read as weather effects. Within cities the annual sine and cosine alone explain 62–92% of the variance of daily UTCI, 76% on average, so they enter as controls and the thermal coefficient is read with them in the model. Figure 2 pools absolute UTCI across cities, whereas the model targets within-city deviations from each city's

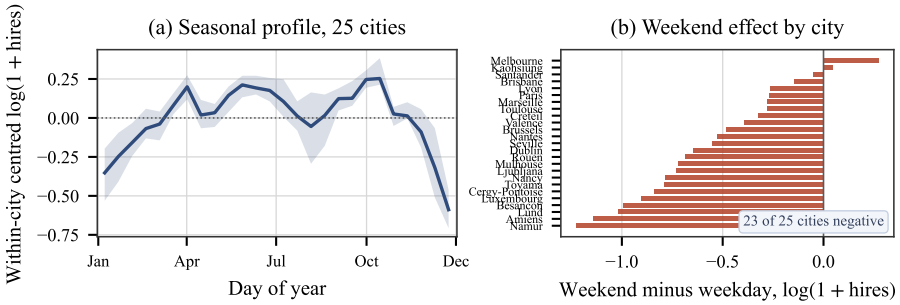


Figure 3: (a) Seasonal profile of within-city centred $\log(1 + \text{hires})$ by day of year, with the same percentile band as Figure 2. (b) Weekend-minus-weekday difference for each primary city, sorted.

own mean conditional on seasonality, so the descriptive peak motivates a nonlinear thermal term rather than estimating an optimum temperature.

4 Plan for the final project

Model. Let U_{it} be estimated hires for city i on local day t and $y_{it} = \log(1 + U_{it})$. The response follows a Student-t observation distribution,

$$y_{it} \sim t_\nu(\mu_{it}, \sigma_i)\,, \qquad \mu_{it} = \alpha_i + \beta_{1i}T_{it} + \beta_{2i}Q_{it} + \beta_{3i}R_{it} + \beta_{4i}W_{it} + \beta_{5i}S_{it} + \beta_{6i}C_{it},$$

where T_{it} is the daily mean UTCI minus the city mean in units of $10°$C, Q_{it} is T_{it}^2 centred within city, R_{it} is $\log(1 + \text{mm})$ minus the city mean, scaled by its within-city standard deviation, W_{it} is a weekend indicator, and S_{it} and C_{it} are the annual sine and cosine of $2\pi d_{it}/365.25$ with d_{it} the day of year. Both weather terms carry the within-city deviation; the city's own rainfall level enters the moderator below. The Student-t distribution accommodates the extreme days a Gaussian would follow. Each city parameter comes from a normal population distribution in non-centred form, $\theta_i = \mu_\theta + \tau_\theta z_i$ with $z_i \sim \text{Normal}(0, 1)$. The population means of the thermal and rainfall coefficients are linear in the city's mean UTCI and mean precipitation, which is the direct test for question (3).

Priors. Priors sit on the $\log(1 + U)$ scale: population intercept $\text{Normal}(7, 2)$ with city intercept scale $\text{HalfNormal}(2)$; thermal coefficient per $10°$C $\text{Normal}(0, 0.5)$; quadratic $\text{Normal}(0, 0.3)$; within-city rainfall $\text{Normal}(0, 0.3)$; weekend $\text{Normal}(0, 0.4)$; seasonal $\text{Normal}(0, 0.8)$; moderators $\text{Normal}(0, 0.2)$ to $\text{Normal}(0, 0.3)$; and city scales HalfNormal matched to the population scale. Degrees of freedom are $\nu = 2 + \text{Exponential}(0.1)$. A prior predictive check will verify the implied range of daily hires before the outcomes are examined.

Inference and checks. Posterior inference uses PyMC 6.3.2 [Abril-Pla et al., 2023] with the nutpie NUTS sampler [Seyboldt, 2024]: four chains, 2,000 warmup and 2,000 retained draws each, seed 27 and `target_accept` $= 0.92$. Convergence is judged by rank-normalised $\widehat{R}$ [Vehtari et al., 2021], bulk and tail effective sample size, Monte Carlo standard error, divergences and tree depth, with $\widehat{R} \leq 1.01$ and both ESS measures at 400 as thresholds. Assessment follows the Bayesian workflow of Gabry et al. [2019]: prior and posterior predictive checks on daily means, standard deviations, quantiles and extreme values; an eight-week hold-out per city scored by interval coverage, mean absolute error and log predictive density; and residual checks. Four sensitivity analyses are planned: the full 29-city panel, weather prior scales multiplied by 0.5 and 2, a Gaussian observation distribution, and a weather-only specification without the seasonal terms. Reported quantities will be the thermal and rainfall percentage changes, the city-level spread of both coefficients, and the moderator coefficients that answer question (3).

Scope. Each city contributes one year, so study-period averages stand in for climate normals, and hires are inventory-change estimates that may include rebalancing moves; system size, pricing and operating policy are unobserved. Conclusions are limited to the 25 continuous dock-based systems.

Contribution statement

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